P2. Gestiune laboratoare studenti

Iteration 1 - IMPLEMENTED

Features

- Student management with CRUD operations
- Problem management with CRUD operations
- CLI interface with help and search functionality

Data Models

- `Student` class with id, name, group properties
- `Problem` class with lab_number, problem_number, description, deadline
- `StudentRepository` for student storage and operations
- `ProblemRepository` for problem storage and operations
- `StudentService` for business logic layer

CLI Commands Implemented

Student Operations:

- `add_student <name> <group>` - Add new student
- `remove_student <student_id>` - Remove student by ID
- `list_students` - List all students
- `search_student <type> <term>` - Search by name, id, or group

Problem Operations:

- `add_problem <lab_problem> <description> <deadline>` - Add new problem
- `remove_problem <lab_problem>` - Remove problem by ID
- `list_problems` - List all problems
- `search_problem <lab_problem_id>` - Search problem by ID

Utility Commands:

- `help` - Show command help
- `clear` - Clear screen
- `exit/quit` - Exit application

Testing

Test Coverage:

- Student model: Constructor, getters, setters
- Student repository: Add, remove, modify, search operations
- Student service: Business logic validation
- Problem model: Constructor, getters, setters
- Problem repository: Add, remove, modify, search operations

Test Execution: Run `python3 test_runner.py` to execute all tests.

Test Cases Include:

- Basic CRUD operations for students and problems
- Search functionality validation
- Error handling for invalid IDs
- Edge cases like empty repositories
- Data integrity and validation

Iteration 2 - PLANNED

Tasks: Assignments

- `AssignmentRepository` for linking students to problems
- CLI for `create_assignment <student_id> <problem_id>`
- CLI for `grade_assignment <assignment_id> <grade>`
- Assignment search and listing functionality

Iteration 3 - PLANNED

Tasks: Statistics and Reports

- Generate student statistics
- Problem completion reports
- Grade analytics
- Export functionality